

Quantitative Discovery of Qualitative Information: A General Purpose Document Clustering Methodology

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Joint work with Justin Grimmer (Harvard $\rightsquigarrow$ Stanford)

Some context for related technology

- <http://ow.ly/14hDU> (play after ad)
- <http://ow.ly/14h36> (play 10:00-12:06)

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- (We focus on texts, our methods apply more broadly)

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- Its no surprise that automated algorithms can help, but which algorithms?

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- **Deep problem in cluster analysis literature: no way to know which method will work ex ante**

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 - An **organized** list will make the search possible
 - E.g.,: consider two clusterings that differ only because one document (of many) moves from category 5 to 6

Our Idea: Meaning Through Geography

Set of clusterings

Our Idea: Measuring Through Geography

Set of clusterings $\approx$
A list of unconnected addresses

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	195	Car	C
375 566-1282 28 Allen St Ipswich 01938	978 356-9960		
81 447-4101 38 Lowell St 02131	617 323-7639		
90 257-9961 F Russell St 02139	617 442-9780		
375 566-1282 B Had 02136	617 361-5253		
375 566-1282 Justicia 50 Decatur Cha 02129	617 241-0152		
375 566-1282 Lucilla 124 Harvard Cam 02138	617 491-5621		
361-0380 M 95 Howe St 02133	617 323-9713		
375 566-1282 Mehin 501 Green Cam 02139	617 576-1061		
375 566-1282 Corte Nicholas 38 Apollon Boston 02134	617 695-6996		
375 566-1282 Cargena O 4 Bedford St 02138	617 338-9219		
375 566-1282 Carten Thos J Sr & Claire 1 Ivesville Rd 02138	617 698-6163		
375 566-1282 Thomas & Kathleen 50 Thompson Ln 02136	617 696-6919		
375 566-1282 Carter A St 02133	617 329-2257		
375 566-1282 A Harbor 375 566-1282	617 442-5230		
375 566-1282 A 31 Bethune Wy Roxbury 02119	617 442-1219		
375 566-1282 A 200 Putnam Av Cambridge 02142	617 492-4174		
375 566-1282 A M 255 Murchie Av Bos 02118	617 266-7153		
375 566-1282 Adams 381 Centre St 02138	617 698-9074		
375 566-1282 Alice 108 Elmwood St 02133	617 425-0193		
375 566-1282 Alice 40 Market Cambridge 02139	617 945-2711		
375 566-1282 Andrew F 42 Wal St 02133	617 625-7632		
375 566-1282 Carter Anne MD 1161 Beacon St 02144	617 739-1022		
375 566-1282 Carter Adhans 272 Newbury Boston 02134	617 536-6329		
375 566-1282 B E 48 Graduate Av Mt 02154	617 296-6911		
375 566-1282 Carter Barbara L MD Tufts New England Medical Center Bos 02111			
375 566-1282 Carter Becky St 02134	617 636-9051		
375 566-1282 Carter Adhans 617 523-4368			
375 566-1282 Bernard J 122 Cambridge F Bos 02138	617 567-3430		
375 566-1282 Bithiah 25 Medway Dor 02124	617 298-8713		
375 566-1282 Bithiah 25 Medway Dor 02124	617 367-9931		
375 566-1282 Carter Broadcasting Co 26 Park Plt Bos 02134	617 423-9210		
375 566-1282 Carter & Burgess Consultants Inc 73 East St Cam 02141	617 225-0200		
375 566-1282 Carter C 2000 Commonwealth Av Bos 02135	617 782-2118		
375 566-1282 C 219 Concord Av East Boston 02128	617 569-1545		
375 566-1282 C 259 Harvard Cam 02138	617 491-4822		
375 566-1282 C 2108 Mt 02124	617 296-4392		
375 566-1282 C & M 41 Burroughs Jan 02138	617 524-9558		
375 566-1282 Carter F 24 Hillock Bos 02131	617 327-1105		
375 566-1282 Faye & Ricky 207 Columbia Av Bos 02136	617 437-7331		
375 566-1282 Francis S 134 Temple W Av Bos 02132	617 323-6781		
375 566-1282 Franklin & Anne 251 Mt Auburn Cam 02138	617 354-0798		
375 566-1282 Fred 42 Haverhill Jan 02136	617 524-3078		
375 566-1282 Fred 96 Harvard Mt 02138	617 698-1343		
375 566-1282 G & R 8 Myer Den 02134	617 436-8906		
375 566-1282 G T 27 Fyfield Av Sun 02145	617 623-7121		
375 566-1282 Gayle 25 Providence Dor 02124	617 825-0322		
375 566-1282 Geo S 115 Mass Hill Rd Jan 02138	617 522-3215		
375 566-1282 George 125 Nathan Bos 02114	617 367-9548		
375 566-1282 Carter Halliday Associate 107 S Street Bos 02111	617 456-1689		
375 566-1282 Carter Harry F 26 Baum St Rd W Av Bos 02132	617 325-5465		
375 566-1282 Carter Hide Co Inc 144 Sumner Bos 02111	617 542-7987		
375 566-1282 Carter Hilary 41 Harvey Cam 02140	617 876-2750		
375 566-1282 Horace 381 Walnut Av Roxbury 02119	617 442-5307		
375 566-1282 Howard Jr 28 Notre Dame Bos 02118	617 445-5552		
375 566-1282 J Cam 41 Chatham Ave 02144	617 334-2658		
375 566-1282 J 518 Harvard Bos 02144	617 232-7990		
375 566-1282 J 775 Wy Young Plt Roxbury 02132	617 730-9483		
375 566-1282 Carter J Jacques MD 1 Broadline Plt Bos 02144	617 323-5374		
375 566-1282 Carter J M 3410 Columbia St Bos 02137	617 735-8787		
375 566-1282 Carter J M Ornamental Ironworks 3410 Columbia St Bos 02137	617 664-1040		
375 566-1282 Carter Barbara L MD Pembroke Tpk 02174	617 436-5353		
375 566-1282 Carter J Veal Co 40 Newbury St 02138	617 442-1775		
375 566-1282 Carter James 1573 Cambridge St Cam 02138	617 492-1214		
375 566-1282 James 352 Fisher Av Roxbury 02108	617 739-2193		
375 566-1282 James 31 Gold Star Rd Cambridge 02140	617 876-8841		
375 566-1282 Jan L 34 Roslindale Rd Mt 02126	617 361-0773		
375 566-1282 Jan L 34 Roslindale Rd Mt 02126	617 364-0435		
375 566-1282 Jeffrey 40 Warren Av Bos 02134	617 426-5994		
375 566-1282 John 11 Mansfield Rd 02134	617 987-2163		
375 566-1282 John 107 Sumner Bos 02135	617 423-4334		
375 566-1282 John 40 Westwood Rd 02125	617 282-1535		
375 566-1282 June O 129 A Summit Av Bos 02138	617 734-6109		
375 566-1282 K 280 Beverly Av Dorchester 02122	617 265-9456		
375 566-1282 K 17 Exford Dorchester 02122	617 282-1931		
375 566-1282 Carter Nellie E 323 Maudsley Av Bos 02135	617 267-6483		
375 566-1282 Nicholas S F 115 Randolph Av Mt 02136	617 698-5307		
375 566-1282 Nick 21 Gayfield Bos 02114	617 267-5222		
375 566-1282 Nick & Debbi 156 Vernick Rd Newton 02459	617 527-0480		
375 566-1282 Nicole 38 Chickadee Dr 02125	617 822-1203		
375 566-1282 P 40 Cranston Plt Bos 02135	617 427-4754		
375 566-1282 P E 501 E South St Bos 02137	617 268-8213		
375 566-1282 P L 44 Hattings Bos 02131	617 427-9170		
375 566-1282 P P 91 Boyer Jan 02134	617 968-8692		
375 566-1282 Paul & Constance 114 Aspen Av W Bos 02130	617 325-2036		
375 566-1282 Paul E 501 E South St Bos 02137	617 268-4546		
375 566-1282 Paul M 27 Union Plt 02135	617 787-2115		
375 566-1282 Carter Pike Driving Inc 27 Beaver Ct Frankington 02702	617 881-235-0488		
375 566-1282 Carter Prudence 40 Franklin Woburn 02127	617 393-3782		
375 566-1282 Prudence 40 Franklin Woburn 02127	617 926-7063		
375 566-1282 Reginald 106 Broadview Dorchester 02215	617 541-2843		
375 566-1282 Renee & Andrew 10 Walnut Bos 02108	617 720-3765		
375 566-1282 Carter Rice Dowd Baker Dennis Publishing 163 Main Woburn 01887			
375 566-1282 Toll Free-Old 'I' & Then 800 638-1671			
375 566-1282 Cost Rec Industrial Prod 113 Main Woburn			
375 566-1282 Toll Free-Old 'I' & Then 800 619-7447			
375 566-1282 Toll Free-Old 'I' & Then 800 648-7447			
375 566-1282 Headquarters 413 Main Woburn 01887			
375 566-1282 Cal 978 988-7447			
375 566-1282 Ingalls Ocean 163 Main Woburn 01887			
375 566-1282 Toll Free-Old 'I' & Then 800 638-1673			
375 566-1282 Carter Richard 207 Commonwealth Av Brighton 02109	617 987-0836		
375 566-1282 Richard A 87 Mt Vernon Bos 02106	617 566-7293		
375 566-1282 Carter Richard A MD 130 Commonwealth Av Bos 02136	617 267-0710		
375 566-1282 Carter Richard K 13 Merwin St 02137	617 268-9448		
375 566-1282 C 175 Melrose Av Cam 02142	617 864-1535		
375 566-1282 Rogers 150 St Brauns Bos 02131	617 424-6148		
375 566-1282 Roy 41 Concord Av 02124	617 491-6115		
375 566-1282 Royce 18 Sumner Cha 02129	617 241-0418		

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90 257-9961	Cartagena Aveth	1 Fairview Rd 02139	617 442-9780	
37 566-1282	Cartagena Aveth	1 Fairview Rd 02139	617 442-9780	
37 566-1282	Cartagena Aveth	1 Fairview Rd 02139	617 442-9780	
361-0380	Carte Nicholas	38 Avenue Boston 02114	617 695-6996	
37 566-4548	Carte Nicholas	38 Avenue Boston 02114	617 695-6996	
37 628-8248	Carten Thos J Sr & Claire	1 Lavalley Rd 02131	617 338-9219	
37 445-5116	Carten Thos J Sr & Claire	1 Lavalley Rd 02131	617 338-9219	
37 822-9902	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 427-5712	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 569-2698	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 667-5190	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 569-1417	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 822-9902	Carte A Inc	50 Thompson Ln 02136	617 696-6919	
37 296-1593	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 670-2078	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 623-9001	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 296-4725	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 542-1521	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 364-5232	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 541-5649	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 739-2662	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 879-0030	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 541-3948	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 836-1511	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 569-4119	Carte Anne MD	1101 Beacon St 02464	617 739-1022	
37 800-4782	Carte Anne MD	1101 Beacon St 02464	617 739-1022	

C



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90 257-9961	Cortagema Aveth	1 Fairview Rd 02139	617 442-9780		
37 566-1282	Jessica	8 Had 02136	617 361-5253		
37 564-5188	Lucilla	124 Harvard Cam 02129	617 241-0152		
361-0380	M 95 Howe Box 02133	617 491-5621			
	M 95 Howe Box 02133	617 323-9713			
	M 95 Howe Box 02139	617 576-1061			
	Carte Nicholas	38 Appleton Boston 02114	617 695-6996		
37 566-4548	Carlegeno O 448ford Box 02131	617 338-0219			
37 628-8248	Carten Thos J Sr & Claire	1 Ivesdale Rd 02136	617 698-6163		
37 445-5116	Thomas & Kathleen	50 Thompson Ln 02136	617 696-6919		
37 822-9992	Carte A Box 02133	617 339-2257			
37 427-5712	A Huber	617 442-5230			
37 569-2698	A 31 Bethune Wy Rindley 02131	617 442-1219			
37 667-5190	A 200 Putnam Av Cambridge 02140	617 492-4174			
37 569-1417	Adams	301 Centre St 02136	617 698-7074		
37 338-9110	Allice	208 Elmwood Rd 02133	617 425-0193		
	Allice	40 Market Cambridge 02139	617 945-2711		
	Andrew F 42 Wal St 02133	617 625-6253			
37 825-1919	Carte Anne MD	617 739-1022			
37 296-1593	1161 Beacon W 02144	617 336-6329			
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37 364-5232	Bithiah	25 Midway Dr 02124	617 298-8713		
37 541-5649	Bishop B 26 Newbury St 02138	617 367-9931			
37 739-2662	Cartier Broadcasting Co	26 Park Pl 02136	617 423-0210		
37 879-0030	Cartier C 2000 Cambridge Av 02135	617 225-0200			
37 936-1511	C 219 Harvard Av East Boston 02128	617 782-2118			
37 569-4119	C 109 Harvard Cam 02138	617 491-4822			
37 569-4782	C & M 41 Burroughs Jan 02136	617 524-5594			
	Carter F 24 Hilltop Box 02131	617 327-1105			
	Faye & Ricky	107 Columbia Av Box 02136	617 437-7331		
	Francis S 134 Temple W Av 02132	617 323-6781			
	Franklin & Anne	291 Mt Auburn Cam 02138	617 354-0798		
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	G & B 8 Harvard Box 02134	617 436-8906			
	G T 27 Franklin Av 02135	617 623-7121			
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	Geo S 115 Miss Hill Rd Jan 02136	617 522-3215			
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	Carter Halliday Associate	107 S Street Box 02111	617 456-1689		
	Carter Harry F	107 S Street Box 02111	617 325-5465		
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	Carter Hilary 41 Harvey Cam 02140	617 876-2750			
	301 Walnut Av Rindley 02131	617 442-5307			
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	J 31 Chatham Box 02144	617 334-2658			
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	J 775 The Weymouth Wy 02135	617 323-5574			
	Carter J Jacques MD	617 735-8787			
	Carter J M	1 Broadview Pl Box 02144	617 735-8787		
	3410 Columbia Rd S Box 02137	617 464-1040			
	Carter J M Ornamental Ironworks	Prudential Tolls 617 436-5353			
	Carter J Veal Co	40 Newbury St 02138	617 442-1775		
	Carter James	1573 Cambridge St Cam 02136	617 492-1214		
	James 102 Fisher Av Rindley 02136	617 739-2193			
	J 31 Gold Star Rd Cambridge 02140	617 876-8841			
	Jas L 34 Broadview Rd 02136	617 361-0773			
	James 14 Adams Rd Newton 02458	617 564-0435			
	John 41 Warren Av Box 02136	617 426-5994			
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	K 109 Weymouth Wy 02135	617 265-9456			
	K 17 Diamond Drive 02123	617 282-1593			
	Carter Nellie E	323 Mauchess Av Box 02135	617 267-6483		
	Nicholas S F	115 Randolph Av 02136	617 698-5307		
	Nick 21 Fairfield Box 02114	617 267-5222			
	Nick & Debbi	156 Vermont Rd Newton 02459	617 527-0480		
	Norman G	38 Chickadee Dr 02125	617 822-1203		
	P 41 Cranford Pl Box 02135	617 427-4754			
	P E 501 E South S Box 02137	617 268-4213			
	P L 44 Matthews Box 02131	617 427-9170			
	P R 81 Boyer Jan 02136	617 983-8692			
	Paul & Constance	114 Adams Av W Box 02130	617 325-2036		
	Paul E 501 E South S Box 02137	617 268-4546			
	Paul M 27 Union St 02135	617 787-2115			
	Carter Pile Driving Inc 17 Beaver Ct	Tramington 02102	617 781-235-0488		
	Carter Prudence	40 Franklin Waterbury 02172	617 393-3782		
	Prudence	40 Franklin Waterbury 02172	617 393-3782		
	Reginald	100 Broadview Cambridge 02215	617 541-2843		
	Renee & Andrew	100 Broadview Cambridge 02215	617 541-2843		
	Carter Rice Dowd	341 Main Waterbury 02172	617 720-3765		
	Toll Free 800-1-800-838-1671				
	Toll Free 800-1-800-616-7447				
	Toll Free 800-1-800-616-7447				
	Headquarters 412 Main Waterbury 02172				
	Call	978 988-7447			
	Inglis Crane 303 Main Waterbury 01887	800 638-1673			
	Toll Free 800-1-800-616-7447				
	Carter Richard	207 Carleton Av Brighton 02135	617 987-0836		
	Richard A 9748 Vernon Box 02136	617 566-7293			
	Carter Richard A MD	170 Centre St Box 02136	617 267-0710		
	Carter Richard M	133 Merwin St Box 02137	617 268-9448		
	Robert L 375 Madison Av Cam 02140	617 864-1535			
	Roger 130 St Pauls Box 02131	617 424-6148			
	Roy 41 Concord Av 02138	617 491-6115			
	Royce 185 Salisbury Box 02129	617 241-9418			



$\approx$ We develop a (conceptual) geography of clusterings

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Make it easy to choose best clustering from millions of choices

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You choose one (or more), based on insight, discovery, useful information,...



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- Meila (2007): derives same metric using different axioms (lattice theory)

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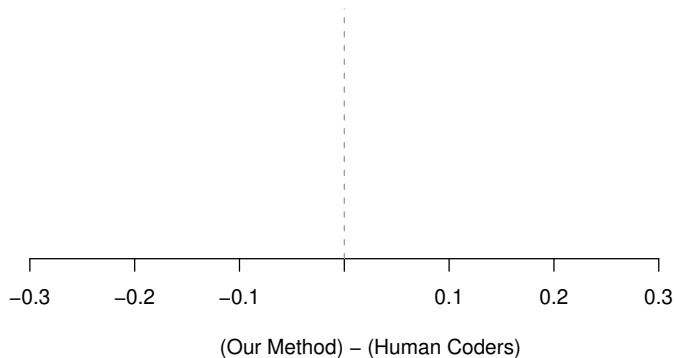
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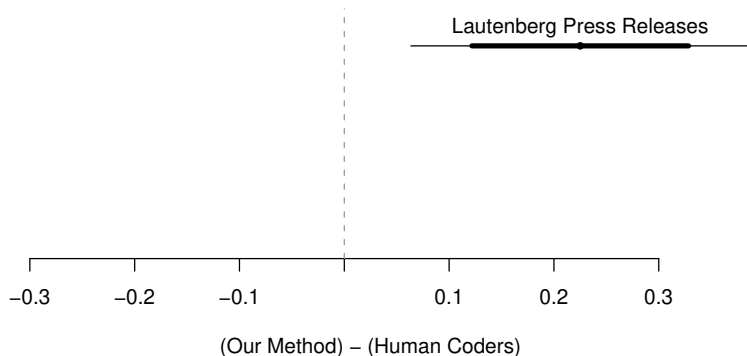
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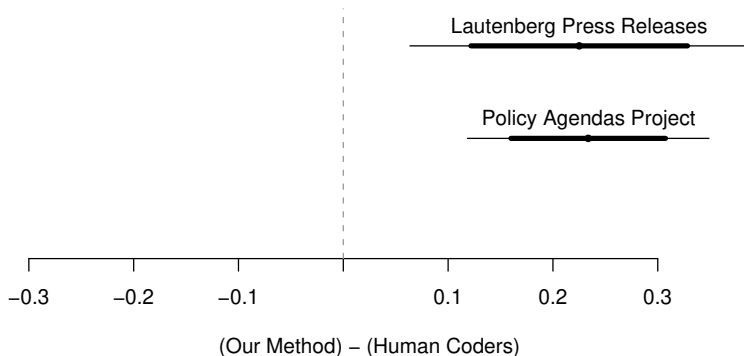


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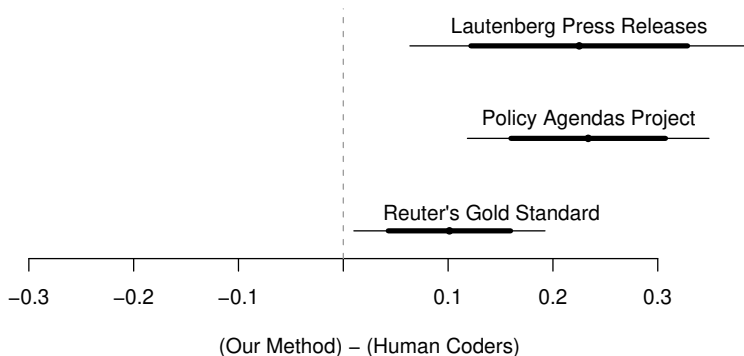
Lautenberg: 200 Senate Press Releases (appropriations, economy, education, tax, veterans, ...)

Evaluation 1: Cluster Quality



Policy Agendas: 213 quasi-sentences from Bush's State of the Union (agriculture, banking & commerce, civil rights/liberties, defense, ...)

Evaluation 1: Cluster Quality



Reuter's: financial news (trade, earnings, copper, gold, coffee, . . .); "gold standard" for supervised learning studies

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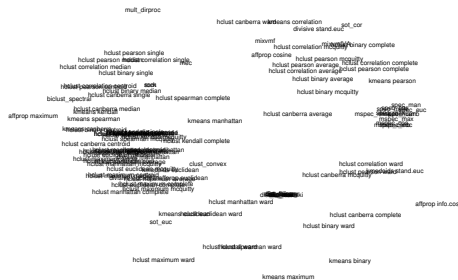
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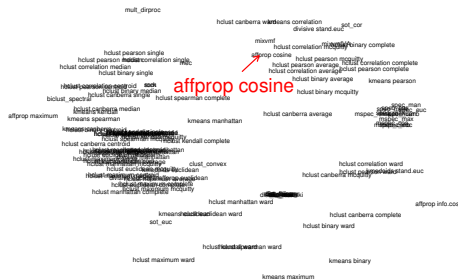
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Example Discovery

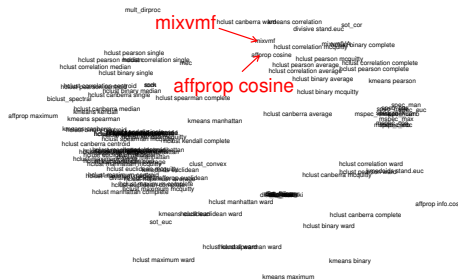


Example Discovery



Red point: a **clustering** by Affinity Propagation-Cosine (Dueck and Frey 2007)

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Close to:

Mixture of von Mises-Fisher distributions (Banerjee et. al. 2005)

Space between methods:



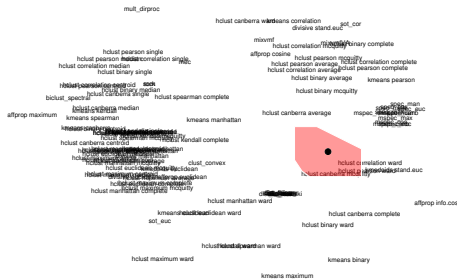
Example Discovery



Space between methods:
local cluster ensemble

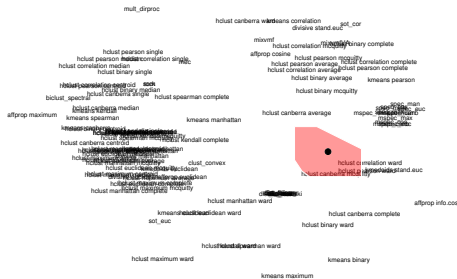


Example Discovery



Mixture:

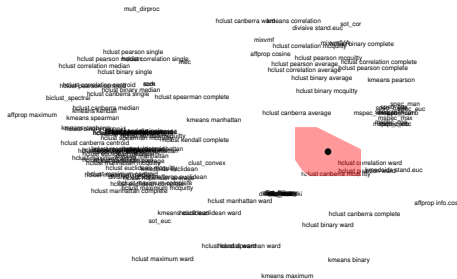
Example Discovery



Mixture:

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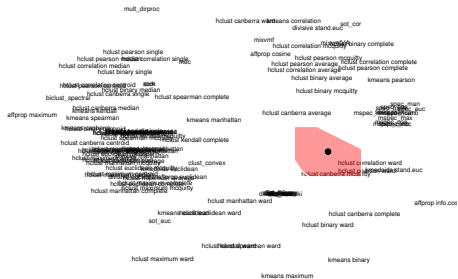


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Random Walk
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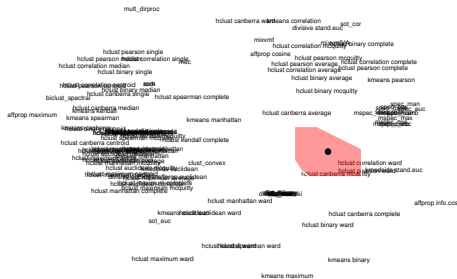
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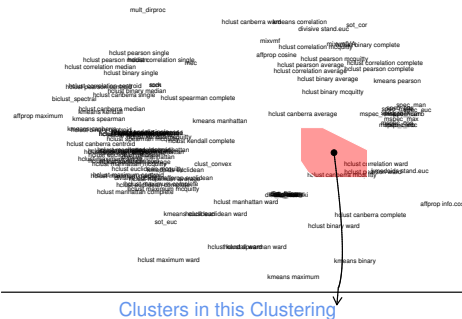
0.04 Spectral clustering

Symmetric
(Metrics 1-6)



Mayhew

Example Discovery

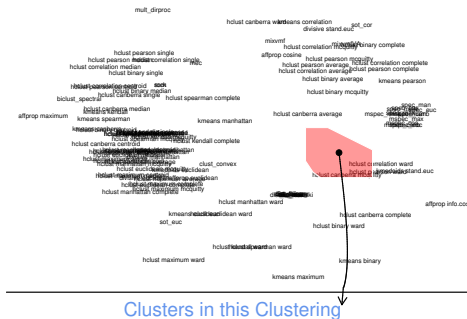


Credit Claiming
Pork

Credit Claiming, Pork:
 “Sens. Frank R. Lautenberg (D-NJ) and Robert Menendez (D-NJ) announced that the U.S. Department of Commerce has awarded a \$100,000 grant to the South Jersey Economic Development District”

Mayhew

Example Discovery



Credit Claiming, Legislation:

"As the Senate begins its recess, Senator Frank Lautenberg today pointed to a string of victories in Congress on his legislative agenda during this work period"

Credit Claiming Pork

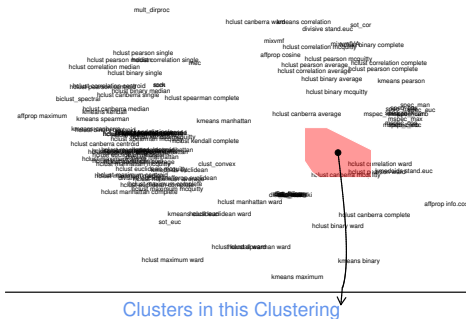
Mayhew

Credit Claiming Legislation

Gary King (Harvard IQSS)

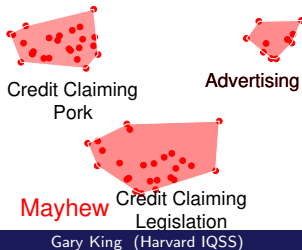
Quantitative Discovery

Example Discovery

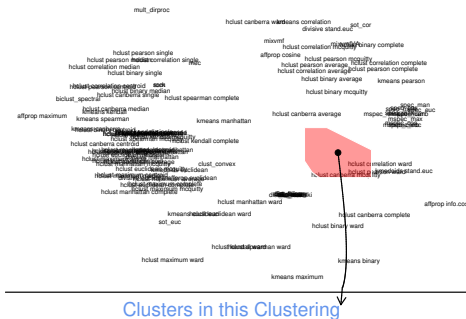


Advertising:

“Senate Adopts
Lautenberg/Menendez Resolution
Honoring Spelling Bee Champion
from New Jersey”

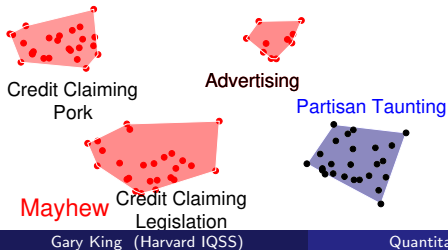


Example Discovery: Partisan Taunting

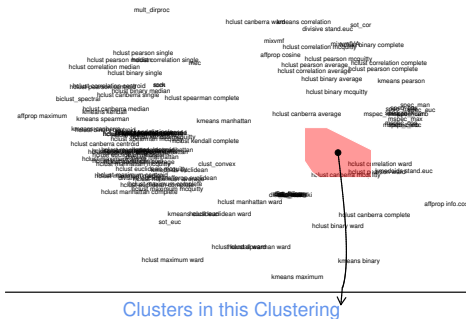


Partisan Taunting:

“Republicans Selling Out Nation on Chemical Plant Security”



Example Discovery: Partisan Taunting



Partisan Taunting:

“Senator Lautenberg’s amendment would change the name of ...the Republican bill...to ‘More Tax Breaks for the Rich and More Debt for Our Grandchildren Deficit Expansion Reconciliation Act of 2006’ ”



Credit Claiming
Pork

Advertising

Partisan Taunting

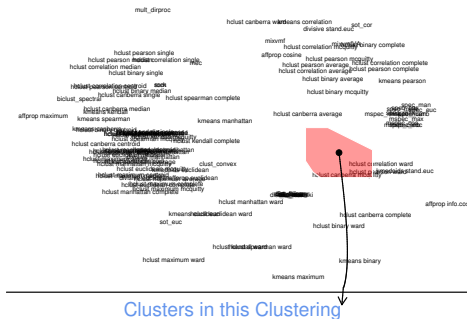
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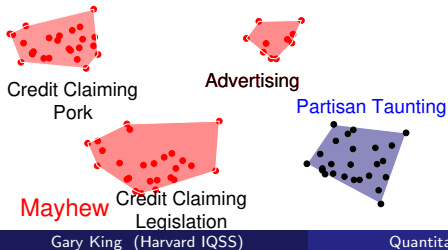
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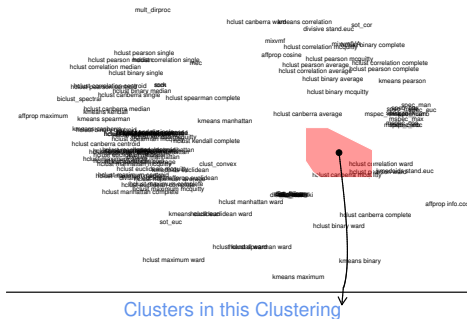
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Definition: Explicit, public, and negative attacks on another political party or its members

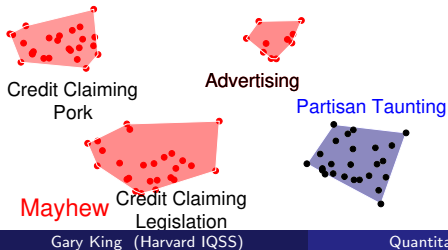


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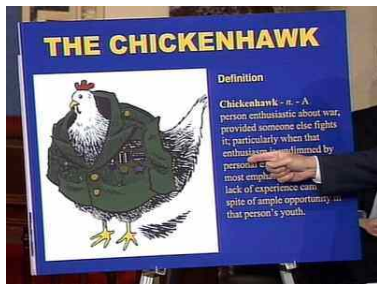
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Taunting ruins deliberation



In Sample Illustration of Partisan Taunting

Taunting ruins deliberation

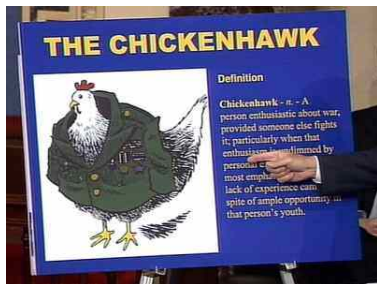


Sen. Lautenberg
on Senate Floor
4/29/04

- “Senator Lautenberg Blasts Republicans as ‘Chicken Hawks’ ”
[Government Oversight]

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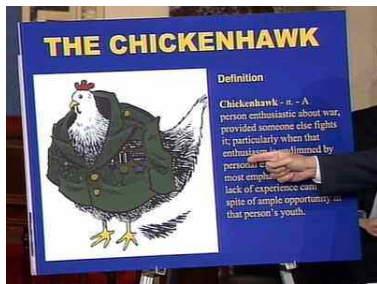


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- "The scopes trial took place in 1925. Sadly, President Bush's veto today shows that we haven't progressed much since then" [Healthcare]
- "Every day the House Republicans dragged this out was a day that made our communities less safe." [Homeland Security]

Out of Sample Confirmation of Partisan Taunting

- Discovered using 200 press releases; 1 senator.

Out of Sample Confirmation of Partisan Taunting

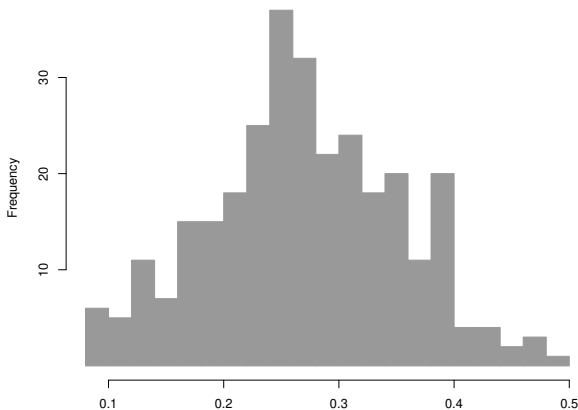
- Discovered using 200 press releases; 1 senator.
- Confirmed using 64,033 press releases; 301 senator-years.

Out of Sample Confirmation of Partisan Taunting

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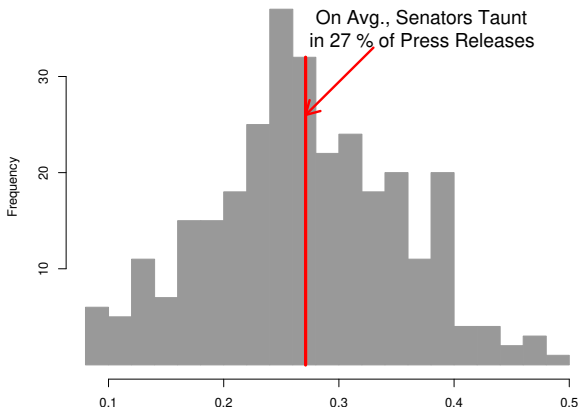
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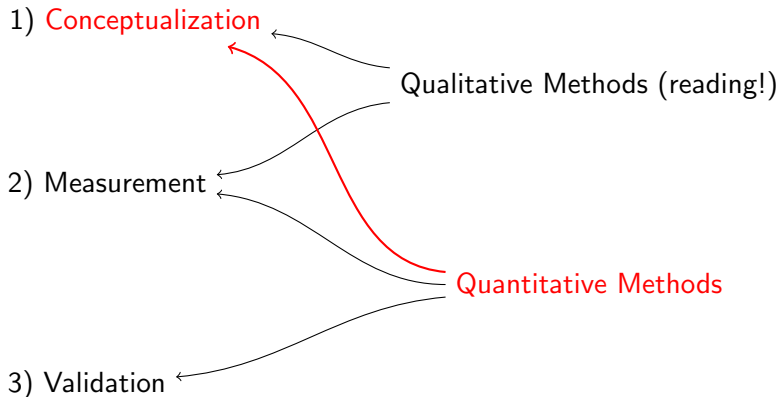


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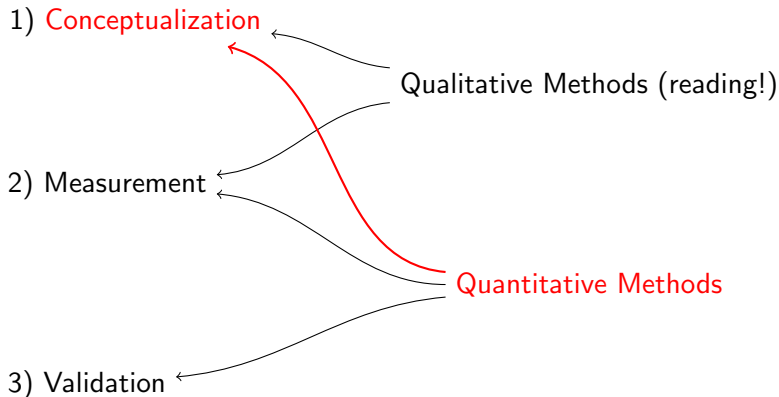


Advancing the Objective of Discovery



Quantitative methods for conceptualization: aiding **discovery**

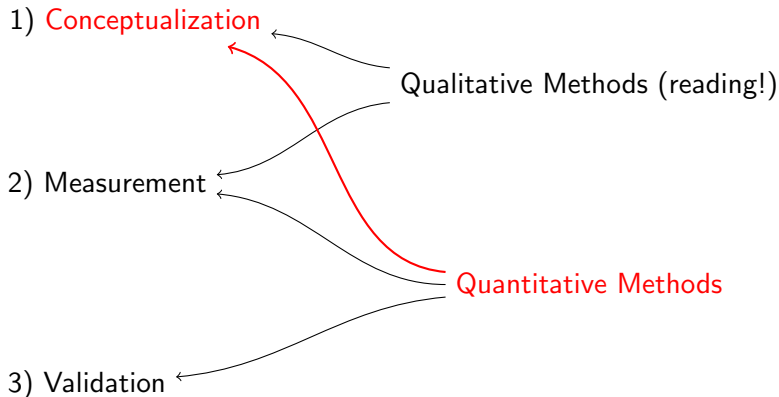
Advancing the Objective of Discovery



Quantitative methods for conceptualization: aiding **discovery**

- Few formal methods designed explicitly for conceptualization

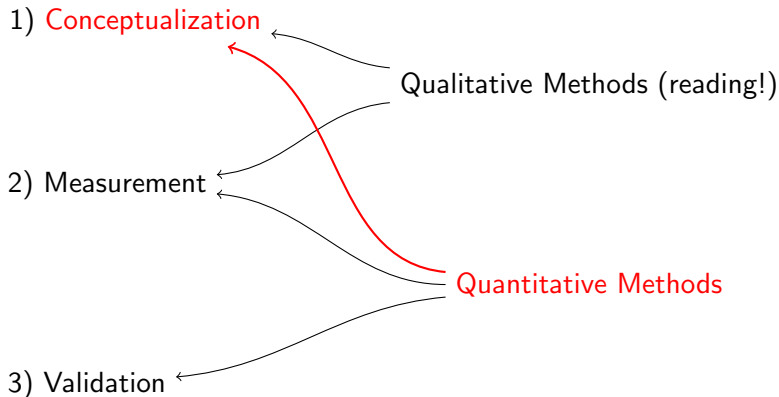
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Advancing the Objective of Discovery



Quantitative methods for conceptualization: aiding **discovery**

- Few formal methods designed explicitly for conceptualization
- **Belittled**: “Tom Swift and His Electric Factor Analysis Machine” (Armstrong 1967)
- Evaluation methods measure progress in discovery

For more information (on adding zooming out to the human ability to zoom in)

<http://GKing.Harvard.edu>